

Team Details

Team Name:

Niriksh

SR. NO	ROLE	NAME	ACADEMIC YEAR
1	Team Leader	Prince Suman	4th
2	Member 1	Chandrahaas	4th
3	Member 2	Aniket Goyal	3rd
4	Member 3	Harsh Mantri	4th

i A team can have up to 4 members including the team leader. Add rows if necessary.

COLLEGE NAME

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Problem Statement Addressed



Edge-AI Defect Classification for Semiconductor Images

DESCRIPTION / DETAILS

Develop an Edge-AI capable system that can automatically detect and classify defects in semiconductor wafer and die images using AI/ML techniques. The solution must operate reliably in real time on low-power edge hardware, reflecting the practical constraints of semiconductor manufacturing environments.

Idea Description



The goal is to develop an edge-ready AI model that automatically classifies wafer defect patterns from wafer SEM images by learning the spatial distribution of defects, enabling fast, scalable, and low-latency semiconductor inspection.

KEY CONCEPT & APPROACH

The approach focuses on developing a computationally efficient, deployable AI-based solution capable of automatically classifying wafer SEM images into predefined defect pattern categories with high accuracy while being suitable for edge deployment.

SOLUTION OVERVIEW

We propose an edge-AI-based wafer defect pattern classification system that automatically analyzes wafer inspection maps and identifies different types of defect patterns for accurate and efficient categorization.

Proposed Solution



We implemented a MobileNet-V3–based deep learning model and trained it using Quantization Aware Training (QAT) to ensure minimal accuracy loss when deploying on low-power edge hardware. Model training was performed on Google Collab using an NVIDIA T4 GPU, allowing efficient handling of the 7000+ image dataset and faster convergence. The training process was conducted in stages, beginning with top-layer training followed by full network fine-tuning, and supported by callbacks such as early stopping to prevent overfitting and optimize training time. After training, the model was converted to INT8 precision, significantly reducing memory footprint, computational complexity, and inference latency. These optimizations enable real-time defect classification directly at the edge while meeting the power and performance constraints of semiconductor manufacturing environments.

Innovation and Uniqueness



KEY INNOVATION

- *Combines semiconductor inspection knowledge with an edge-deployable AI pipeline tailored for real fab constraints.*
- *Uses training-time optimization techniques to ensure the model remains accurate even after aggressive compression.*
- *Introduces a real-time automated inspection approach that can operate continuously on production lines.*
- *Focuses on practical deployability, not just model accuracy in lab conditions.*

Uniqueness

- *Built a domain-specific dataset with realistic variations (brightness, contrast, rotation) to mimic real inspection scenarios with 7000+ images (overall).*
- *End-to-end workflow — from dataset creation to deployment-ready quantized model — developed entirely in-house.*
- *Designed specifically for low memory (approx. 7MB) and accuracy greater than 96.*
- *Architecture allows easy adaptation to new defect categories without redesigning the full system.*

Impact and Benefits



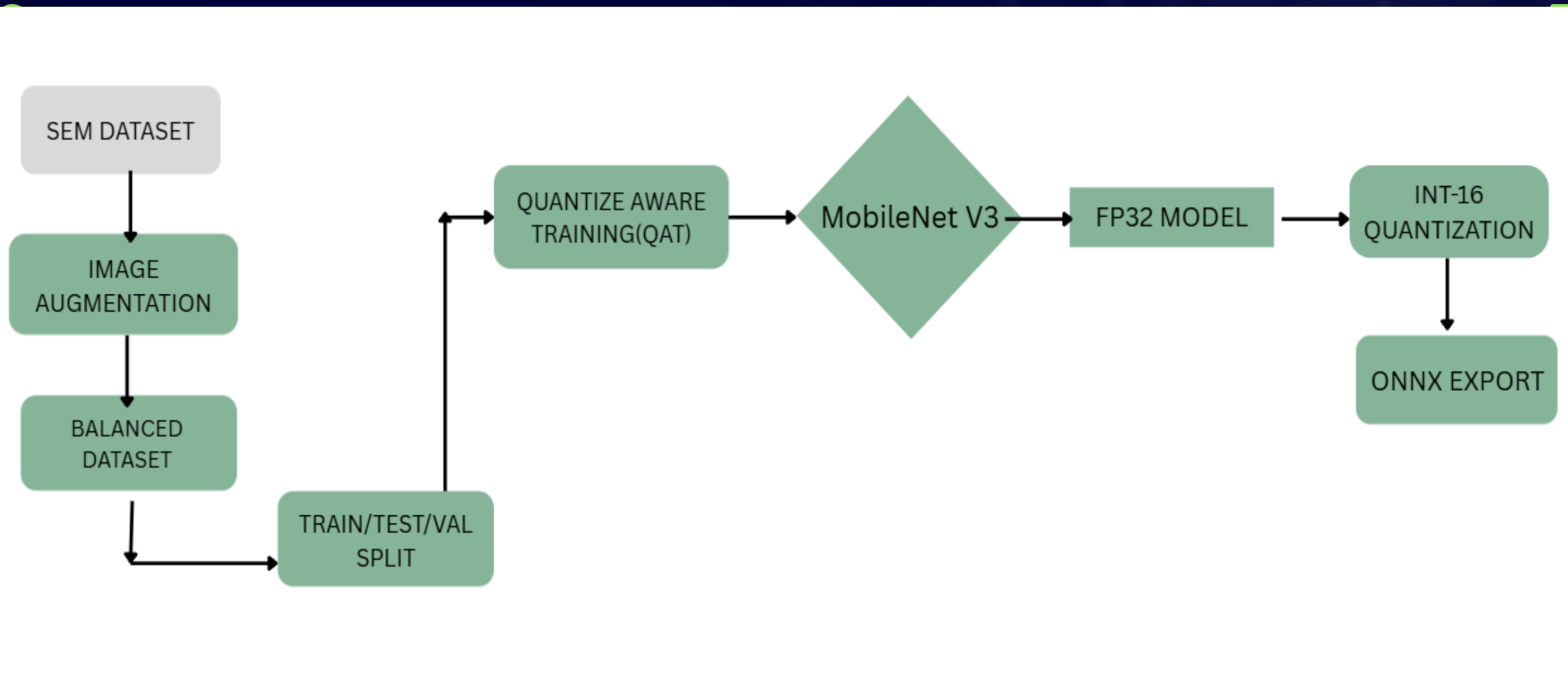
Primary Impact

- *Enables real-time automated detection of semiconductor wafer and die defects.*
- *Eliminates dependency on cloud processing through edge-based AI deployment.*
- *Improves manufacturing efficiency by reducing manual inspection time.*
- *Allows early identification of defects, preventing faulty dies from progressing further.*
- *Enhances overall production reliability and yield.*
- *Supports continuous, on-line inspection in semiconductor fabrication environments.*

Quantifiable Outcomes

- *Faster inspection speed due to real-time edge inference.*
- *Reduced memory and computation requirements using INT8 optimized model.*
- *Lower operational cost by avoiding high-power centralized processing.*
- *Improved classification performance across 7 defect categories.*
- *Scalable deployment across multiple inspection stations.*
- *Increased throughput and reduced rework in fabrication process.*

Overall Flow



Technology & Feasibility/Methodology Used



Describe the technologies, methodologies, or tools you plan to use to implement your idea.

IMPLEMENTATION STRATEGY

- *MobileNet-V3 deep learning architecture used for semiconductor defect classification from wafer and die images.*
- *Quantization Aware Training (QAT) applied and model converted to INT8 for efficient, low-power edge deployment with minimal accuracy loss.*
- *Model developed and trained in Python using TensorFlow/Keras on Google-Colab (NVIDIA T4 GPU) with data augmentation and staged fine-tuning.*



Software Architecture



Hardware Components



Development Tools

GitHub & Dataset Link



GitHub Repository

<https://github.com/chandrahaas1563/Niriksh>



Dataset

https://drive.google.com/drive/folders/1IJQq4K5m4Q3ggibhiM3BFa9Ulwhy0yNn?usp=drive_link

Research and References



Research Background & Methodology

Briefly describe the research foundation or scientific principles supporting your idea.

- *The solution is based on Convolutional Neural Network (CNN) principles for automated feature extraction and defect classification from semiconductor images, combined with model optimization and quantization techniques to enable accurate, real-time inference on low-power edge devices.*



References & Citations

List key papers, articles, or data sources.

<https://arxiv.org/html/2407.10348v1#S3>

<https://arxiv.org/pdf/2506.03345>

<https://learnopencv.com/ensemble-deep-learning-based-defect-classification-and-detection-in-sem-images/#Datasets->